一、严格表述题

1. *Cauchy*叙述：对$\forall \varepsilon>0, \exists \delta_{\varepsilon}>0$,使得当$0<\left|\boldsymbol{x}-\boldsymbol{x}_0\right|<\delta_{\varepsilon}$,都有

$\left|f\left(\boldsymbol{x}\right)-f\left(\boldsymbol{x}_0\right)\right|<\varepsilon$

*Heine*叙述：对$\forall\left\{\boldsymbol{x}_n\right\} \subset D \backslash\left\{\boldsymbol{x}_0\right\}$，满足$\boldsymbol{x}_n \to \boldsymbol{x}_0\left(n \to \infty\right)$，都有

$f\left(\boldsymbol{x}_n\right) \to f\left(\boldsymbol{x}_0\right)\left(n \to \infty\right)$

*Cauchy*收敛原理：对$\forall \varepsilon>0$，$\exists \delta_{\varepsilon}>0$,使得当$\boldsymbol{x}', \boldsymbol{x}'' \in \overset{\circ}{B}_{\delta_{\varepsilon}}\left(\boldsymbol{x}_0\right)$时，都有

$\left|f\left(\boldsymbol{x}'\right)-f\left(\boldsymbol{x}''\right)\right|<\varepsilon$

2. 对向量值映照$\boldsymbol{f}\left(\boldsymbol{x}\right): \mathbb{R}^m \supset D_x \ni \boldsymbol{x} \mapsto \boldsymbol{f}\left(\boldsymbol{x}\right) \in \mathbb{R}^n, \boldsymbol{x}_0$为$D_x$的内点，有

$\boldsymbol{f}\left(\boldsymbol{x}_0+\boldsymbol{h}\right)-\boldsymbol{f}\left(\boldsymbol{x}\right)=\boldsymbol{D f}\left(\boldsymbol{x}_0\right)(\boldsymbol{h})+o\left(|\boldsymbol{h}|_{\mathbb{R}^m}\right) \in \mathbb{R}^n$，$\boldsymbol{D f}\left(\boldsymbol{x}_0\right) \in \mathscr{L}\left(\mathbb{R}^m ; \mathbb{R}^n\right)$,

则称$\boldsymbol{f}\left(\boldsymbol{x}\right)$在$\boldsymbol{x}_0$点可微。

3，第二类曲线积分：设$L$为一条定向光滑曲线，起点为$A$，终点为$B$。在$L$上每一点取切向量$\boldsymbol{\tau}$,使得它与$L$的走向相一致。设

$\boldsymbol{f}\left(x, y, z\right)=P\left(x, y, z\right) \boldsymbol{i}+Q\left(x, y, z\right) \boldsymbol{j}+R\left(x, y, z\right) \boldsymbol{k}$定义在$L$上的向量值函数，则称

$\int_L \boldsymbol{f} \cdot \boldsymbol{\tau} d l=\int_L\left(P\left(x, y, z\right) \cos \alpha+Q\left(x, y, z\right) \cos \beta+R\left(x, y, z\right) \cos \gamma\right) d l$为第二类曲线积分。

第二类曲面积分，同样，有$\boldsymbol{f}\left(x, y, z\right)=P\left(x, y, z\right) \boldsymbol{i}+Q\left(x, y, z\right) \boldsymbol{j}+R\left(x, y, z\right) \boldsymbol{k}$

为定义在定向光滑曲面$\Sigma$上的向量值函数，设曲面上单位法向量$\boldsymbol{n}=\left(\cos \alpha, \cos \beta, \cos \gamma\right)$,

则$\iint_{\Sigma} \boldsymbol{f} \cdot \boldsymbol{n} d S=\iint_{\Sigma}\left(P\left(x, y, z\right) \cos \alpha+Q\left(x, y, z\right) \cos \beta+R\left(x, y, z\right) \cos \gamma\right) d S$为第二类曲面积分

4，若下列二条件的一个满足，则级数$\sum_{n=1}^{\infty} a_n b_n$收敛：

$(1)(Abel)\left|a_n\right|$单调有界,$\sum_{n=1}^{+\infty} b_n$收敛

$(2)(Dirichlet)\left|a_n\right|$单调趋于0,$\left|\sum_{n=1}^{+\infty} b_n\right|$有界

二、简答题

1. 解：左$=\dfrac{\partial y}{\partial t}=-k n^2 \sin n x \cdot e^{-k n^2 t}$

$\dfrac{\partial y}{\partial x}=e^{-k n^2 t} n \cos n x$

右$=k \dfrac{\partial^2 y}{\partial x^2}=-e^{k n^2 t} k n^2 \sin n x$

则有 左=右

2. 解：$f(x,y)=\dfrac{x^3+y^3}{x^2+y}$

令$y=-x^2+\lambda x^p\ (p>0)$

$$\sim\begin{cases}\lambda x^p\left(1+o(1)\right) & 0<p<2\\ (\lambda-1)x^2 & p=2\\ -x^2\left(1+o(1)\right) & p>2\end{cases}$$

$(i)\,0<p<2\quad f(x,y)=\dfrac{x^3+\lambda^3x^{3p}\left(1+o(1)\right)}{\lambda x^p}=\dfrac{1}{\lambda}x^{3-p}+\lambda^2x^{2p}\left(1+o(1)\right)\to 0$

$(ii)\,p=2\qquad f(x,y)=\dfrac{x^3+(\lambda-1)^3x^6}{\lambda x^2}=\dfrac{1}{\lambda}x+\dfrac{(\lambda-1)^3}{\lambda}x^4\to 0$

$(iii)\,p>2\qquad f(x,y)=\dfrac{x^3-x^6\left(1+o(1)\right)}{\lambda x^3}\to\dfrac{1}{\lambda}\neq 0$

则有$f(x,y)$无极限

3，注意到积分：$\iint\limits_{\mathbb{R}^2}e^{-x^2-y^2}dxdy=\int_0^{2\pi}d\theta\int_0^{+\infty}e^{-r^2}rdr=2\pi\bullet\dfrac{1}{2}=\pi$

又：$\iint\limits_{\mathbb{R}^2}e^{-x^2-y^2}dxdy=\int_{-\infty}^{+\infty}e^{-x^2}dx\int_{-\infty}^{+\infty}e^{-y^2}dy=4\left(\int_0^{+\infty}e^{-x^2}dx\right)^2$

即$\int_0^{+\infty}e^{-x^2}dx=\dfrac{\sqrt{\pi}}{2}$

4，$\lim\limits_{n\to\infty}\left|\dfrac{a_{n+1}}{a_n}\right|=\lim\limits_{n\to\infty}\left(1+\dfrac{1}{n}\right)^n=e$，故收敛半径为$\dfrac{1}{e}$

$x=e^{-1}$时，$\dfrac{n^n}{n!}x^n\sim\dfrac{1}{\sqrt{2\pi n}}$，由比较判别法知发散

$x=-e^{-1}$时，$\left|\dfrac{n^n}{n!}x^n\right|\sim\dfrac{1}{\sqrt{2\pi n}}\to 0$，且$\left|\dfrac{a_{n+1}x^{n+1}}{a_nx^n}\right|=\dfrac{1}{e}\left(1+\dfrac{1}{n}\right)^n<1$，

此*Leibniz*级数收敛

三、级数题

1. 解：$\dfrac{u_{n+1}}{u_n}=\dfrac{(n+1)!}{10^{n+1}}\cdot\dfrac{10^n}{n!}=\dfrac{n+1}{10}$，

$\lim\limits_{n\to\infty}\dfrac{u_{n+1}}{u_n}=\lim\limits_{n\to\infty}\dfrac{n+1}{10}=\infty.$

则有级数发散。

2. 解：$f(x)=\frac{1}{x^2+4x+3}=\frac{1}{(x+1)(x+3)}=\frac{1}{2(1+x)}-\frac{1}{2(3+x)}$

$=\frac{1}{4\left(1+\frac{x-1}{2}\right)}-\frac{1}{8\left(1+\frac{x-1}{4}\right)}$

其中$\frac{1}{4\left(1+\frac{x-1}{2}\right)}=\frac{1}{4}\sum_{n=0}^{\infty}\frac{(-1)^n}{2^n}(x-1)^n \quad (-1<x<3)$

$\frac{1}{8\left(1+\frac{x-1}{4}\right)}=\frac{1}{8}\sum_{n=0}^{\infty}\frac{(-1)^n}{4^n}(x-1)^n \qquad (-3<x<5)$

则有

$f(x)=\frac{1}{x^2+4x+3}=\sum_{n=0}^{\infty}(-1)^n\left(\frac{1}{2^{n+2}}-\frac{1}{2^{2n+3}}\right)(x-1)^n \quad (-1<x<3)$

3, 首先对函数进行偶延拓：$\begin{cases}x, & x\in[0,\pi)\\ -x, & x\in[-\pi,0)\end{cases}$

$a_0=\frac{2}{\pi}\int_0^{\pi}xdx=\pi, a_n=\frac{2}{\pi}\int_0^{\pi}x\cos nxdx=2\bullet\frac{(-1)^n-1}{n^2\pi}$

故$f(x)=\frac{\pi}{2}-\frac{4}{\pi}\sum_{n=1}^{\infty}\frac{\cos(2k-1)x}{(2k-1)^2}$

代入$x=0$,立即有：$\sum_{n=1}^{\infty}\frac{1}{(2k-1)^2}=\frac{\pi}{2}\bullet\frac{\pi}{4}=\frac{\pi^2}{8}$

注意到：$\cos x$全部零点为$\pm\frac{2k-1}{2}\pi$，代入即有所求为：$\frac{\pi^2}{8}\bullet\frac{4}{\pi^2}\bullet 2=1$

四、多元函数微分题

1. 解：事物$\{x,y,z\}\in\mathbb{R}^3, s.t. \bar{\boldsymbol{F}}(x,y)=\boldsymbol{F}\left(z,\begin{bmatrix}x\\y\end{bmatrix}(z)\right)=\begin{bmatrix}x+y+z\\x^2+y^2+z^2-1\end{bmatrix}=0\in\mathbb{R}^2$

$0=\boldsymbol{D}\bar{\boldsymbol{F}}=\boldsymbol{D}_z\boldsymbol{F}+\boldsymbol{D}_{\begin{bmatrix}x\\y\end{bmatrix}}\boldsymbol{F}\cdot\boldsymbol{D}\begin{bmatrix}x\\y\end{bmatrix}(z)\in\mathbb{R}^{2\times1}$

$$\boldsymbol{D}\begin{bmatrix}x\\y\end{bmatrix}(z)=-\left(\boldsymbol{D}_{\begin{bmatrix}x\\y\end{bmatrix}}\boldsymbol{F}\right)^{-1}\cdot\boldsymbol{D}_z\boldsymbol{F}$$

$$\begin{bmatrix}\frac{\partial x}{\partial z}\\\frac{\partial y}{\partial z}\end{bmatrix}=-\begin{bmatrix}\frac{\partial F^1}{\partial x}&\frac{\partial F^1}{\partial y}\\\frac{\partial F^2}{\partial x}&\frac{\partial F^2}{\partial y}\end{bmatrix}^{-1}\cdot\begin{bmatrix}\frac{\partial F^1}{\partial z}\\\frac{\partial F^2}{\partial z}\end{bmatrix}$$

$$\begin{bmatrix}\frac{\partial x}{\partial z}\\\frac{\partial y}{\partial z}\end{bmatrix}=-\begin{bmatrix}1&1\\2x&2y\end{bmatrix}^{-1}\cdot\begin{bmatrix}1\\2z\end{bmatrix}$$

$$\frac{\partial x}{\partial z}=-\frac{1}{2(y-x)}\begin{vmatrix}1&1\\2z&2y\end{vmatrix}=-\frac{2(y-z)}{2(y-x)}=-\frac{y-z}{y-x}$$

$$\frac{\partial y}{\partial z}=-\frac{1}{2(y-x)}\begin{vmatrix}1&1\\2x&2z\end{vmatrix}=-\frac{2(z-x)}{2(y-x)}=-\frac{z-x}{y-x}$$

即$\frac{dx}{dz}=-\frac{y-z}{y-x}$ ， $\frac{dy}{dz}=-\frac{z-x}{y-x}$

2. 解：自变量变换$\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x+y \\ \dfrac{y}{x} \end{bmatrix}$

因变量变换$z = wx =: \Phi\left(\begin{bmatrix} x \\ y \end{bmatrix}, w\left(\begin{bmatrix} u \\ v \end{bmatrix}(x,y)\right)\right)$

则有

$$\boldsymbol{D}z(x,y) = \boldsymbol{D}_{\begin{bmatrix} x \\ y \end{bmatrix}}\Phi + \boldsymbol{D}_w\Phi \cdot \boldsymbol{D}w(u,v) \cdot \boldsymbol{D}\begin{bmatrix} u \\ v \end{bmatrix}(x,y)$$

$$\begin{bmatrix} z_x & z_y \end{bmatrix} = \begin{bmatrix} w & 0 \end{bmatrix} + x \cdot \begin{bmatrix} w_u & w_v \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ -\dfrac{y}{x^2} & \dfrac{1}{x} \end{bmatrix}$$

$$z_x = w + xw_u - \frac{y}{x}w_v = \frac{z}{x} + xw_u - \frac{y}{x}w_v =: \Psi\left(\begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} u \\ v \end{bmatrix}, z\right)$$

$$z_y = xw_u + w_v =: \Theta\left(\begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} u \\ v \end{bmatrix}\right)$$

$$(i)\begin{bmatrix} z_{xx} & z_{xy} \end{bmatrix} = \boldsymbol{D}_{\begin{bmatrix} x \\ y \end{bmatrix}}\Psi + \boldsymbol{D}_{\begin{bmatrix} u \\ v \end{bmatrix}}\Psi \cdot \boldsymbol{D}\begin{bmatrix} u \\ v \end{bmatrix}(x,y) + \boldsymbol{D}_z\Psi \cdot \boldsymbol{D}z(x,y)$$

$$= \begin{bmatrix} -\dfrac{z}{x^2} + w_u + \dfrac{y}{x^2}w_v & -\dfrac{w_v}{x} \end{bmatrix}$$

$$+ \begin{bmatrix} xw_{uu} - \dfrac{y}{x}w_{uv} & xw_{uv} - \dfrac{y}{x}w_{vv} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -\dfrac{y}{x^2} & \dfrac{1}{x} \end{bmatrix}$$

$$+ \frac{1}{x}\begin{bmatrix} \dfrac{z}{x} + xw_u - \dfrac{y}{x}w_v & xw_u + w_v \end{bmatrix}$$

$$\begin{cases} z_{xx} = 2w_u + xw_{uu} - \dfrac{2y}{x}w_{uv} + \dfrac{y^2}{x^3}w_{vv} \\ z_{xy} = w_u + xw_{uu} + w_{uv} - \dfrac{y}{x}w_{uv} - \dfrac{y}{x^2}w_{vv} \end{cases}$$

$$(ii)\begin{bmatrix} z_{xy} & z_{yy} \end{bmatrix} = \boldsymbol{D}_{\begin{bmatrix} x \\ y \end{bmatrix}}\Theta + \boldsymbol{D}_{\begin{bmatrix} u \\ v \end{bmatrix}}\Theta \cdot \boldsymbol{D}\begin{bmatrix} u \\ v \end{bmatrix}(x,y)$$

$$= \begin{bmatrix} w_u & 0 \end{bmatrix} + \begin{bmatrix} xw_{uu} + w_{uv} & xw_{uv} + w_{vv} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -\dfrac{y}{x^2} & \dfrac{1}{x} \end{bmatrix}$$

$$\begin{cases} z_{xy} = w_u + xw_{uu} + w_{uv} - \dfrac{y}{x}w_{uv} - \dfrac{y}{x^2}w_{vv} \\ z_{yy} = xw_{uu} + 2w_{uv} + \dfrac{1}{x}w_{vv} \end{cases}$$

代入化简即可得$\left(\dfrac{y^2}{x^3} + \dfrac{2y}{x^2} + \dfrac{1}{x}\right)w_{vv} = 0$

$(x+y)^2 w_{vv}=0$即$u^2 w_{vv}=0$

五，多元函数积分题

1，本积分为第二类曲面积分，将$Z=\sqrt{x^2+y^2}$直接向$xoy$面投影，立即有：

$$Z_x=\frac{x}{\sqrt{x^2+y^2}},Z_y=\frac{y}{\sqrt{x^2+y^2}},$$

原式为：$-\iint\limits_{D_{xy}}\frac{-x}{\sqrt{x^2+y^2}}\left[x\left[f(x)+g(y)\right]+2x-y\right]+\left[y\left[f(x)+g(y)\right]+2y+x\right]\frac{-y}{\sqrt{x^2+y^2}}$

$+\sqrt{x^2+y^2}\left[f(x)+g(y)+1\right]dxdy$

$=\iint\limits_{D_{xy}}\sqrt{x^2+y^2}dxdy=\int_0^{2\pi}d\theta\int_2^4 r\bullet rdr=\frac{112}{3}\pi$

2,容易知道，截面是一个等边三角形，边长为$\sqrt{2}$，$\boldsymbol{n}=\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$

由*Stocks*公式：$\oint\limits_{\partial S}\boldsymbol{f}\bullet\boldsymbol{\tau}dl=\iint\limits_S(\nabla\times\boldsymbol{a})\bullet\boldsymbol{n}ds$,则原式为：$\iint\limits_S\frac{1}{\sqrt{3}}(-2y-2z-2z-2x-2x-2y)ds$

$=-\frac{4}{\sqrt{3}}\iint\limits_S(x+y+z)ds=-\frac{4}{\sqrt{3}}\iint\limits_S ds=-\frac{4}{\sqrt{3}}\bullet\frac{\sqrt{3}}{4}\bullet 2=-2$

3,引入广义极坐标：$\begin{cases}x=2r\cos\theta\\ y=3r\sin\theta\end{cases},\frac{\partial(r,\theta)}{\partial(x,y)}=6r$

曲线方程为：$r^2=3\sin 2\theta,$

由于图线具有对称性，所以仅计算第一象限部分即可。

$D=\left\{(r,\theta)\middle|\theta\in\left[0,\frac{\pi}{2}\right],r\in\left[0,\sqrt{3\sin 2\theta}\right]\right\}$

所求面积为：$2\iint\limits_D 6r\mathrm{d}r\mathrm{d}\theta=12\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{\sqrt{3\sin 2\theta}}rdr=18$

六、证明题

1. 证明: 设$L(x,y,z;\lambda)=\ln x+2\ln y+3\ln z+\lambda\left(x^2+y^2+z^2-6R^2\right)$

$$(i)\begin{cases}L_x=\dfrac{1}{x}+2\lambda x=0\\ L_y=\dfrac{2}{y}+2\lambda y=0\\ L_z=\dfrac{3}{z}+2\lambda z=0\\ x^2+y^2+z^2=6R^2\end{cases}\Rightarrow\begin{cases}\lambda=-\dfrac{1}{2R^2}\\ x=R\\ y=\sqrt{2}R\\ z=\sqrt{3}R\end{cases}$$

$$(ii)\,\boldsymbol{H}_L=\begin{bmatrix}L_{xx}&L_{xy}&L_{xz}\\ L_{yx}&L_{yy}&L_{yz}\\ L_{zx}&L_{zy}&L_{zz}\end{bmatrix}=\begin{bmatrix}-\dfrac{1}{x^2}+2\lambda&0&0\\ 0&-\dfrac{2}{y^2}+2\lambda&0\\ 0&0&-\dfrac{3}{z^2}+2\lambda\end{bmatrix}=\begin{bmatrix}-\dfrac{2}{R^2}&0&0\\ 0&-\dfrac{2}{R^2}&0\\ 0&0&-\dfrac{2}{R^2}\end{bmatrix}$$

*Hesse*阵为负定矩阵

即有$\left(R,\sqrt{2}R,\sqrt{3}R\right)^T$处为严格极大值点

$$\ln x+2\ln y+3\ln z\le\ln(R)+2\ln\left(\sqrt{2}R\right)+3\ln\left(\sqrt{3}R\right)$$

$$\ln\left(xy^2z^3\right)\le\ln\left(R\cdot2R^2\cdot3\sqrt{3}R^3\right)=\ln\left(6\sqrt{3}R^6\right)$$

$$xy^2z^3\le6\sqrt{3}R^6$$

$$x^2y^4z^6\le108R^{12}=108\left(\frac{x^2+y^2+z^2}{6}\right)^6$$

令$a=x^2,b=y^2,c=z^2$

则有$ab^2c^3\le108\left(\dfrac{a+b+c}{6}\right)^6$

命题得证

2,注意到$(0,0)$为区域内奇点，设$L$围成区域为$\Sigma$，构造$\Sigma_\varepsilon:4x^2+y^2\le\varepsilon^2$，边界为$L_\varepsilon$

在区域$\Sigma-\Sigma_\varepsilon$上运用格林公式:

$$\oint_{L-L_\varepsilon}\frac{(4x-y)dx+(x+y)dy}{4x^2+y^2}=\iint_{\Sigma-\Sigma_\varepsilon}\frac{-2y(4x-y)-\left(4x^2+y^2\right)-\left(4x^2+y^2\right)+(x+y)8x}{\left(4x^2+y^2\right)^2}ds=0$$

故有原式为:$\oint_{L_\varepsilon}\dfrac{(4x-y)dx+(x+y)dy}{4x^2+y^2}=\oint_{L_\varepsilon}\dfrac{(4x-y)dx+(x+y)dy}{\varepsilon^2}$

此时，区域无奇点，再使用格林公式:$\dfrac{1}{\varepsilon^2}\iint_{\Sigma_\varepsilon}1\text{-}(-1)\,dxdy=\dfrac{2}{\varepsilon^2}\pi\cdot\varepsilon\cdot\dfrac{\varepsilon}{2}=\pi$